

List Removals

Time limit: 1.00 s

Memory limit: 512 MB

You are given a list consisting of n integers. Your task is to remove elements from the list at given positions, and report the removed elements.

Input

The first input line has an integer n : the initial size of the list. During the process, the elements are numbered $1, 2, \dots, k$ where k is the current size of the list.

The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the list.

The last line has n integers $p_1, p_2, \dots, p_n$: the positions of the elements to be removed.

Output

Print the elements in the order they are removed.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq p_i \leq n - i + 1$

Example

Input	Output
5 2 6 1 4 2 3 1 3 1 1	1 2 2 6 4

Explanation: The contents of the list are $[2, 6, 1, 4, 2]$, $[2, 6, 4, 2]$, $[6, 4, 2]$, $[6, 4]$, $[4]$ and $[]$.